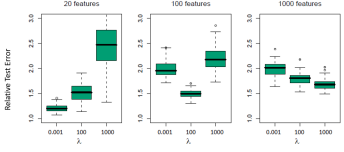
# Introduction to Machine Learning

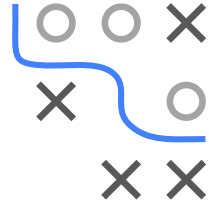
## Feature Selection
## Feature Selection: Motivating Examples




**Learning goals**

- Understand the practical importance of feature selection
- Understand that models with integrated selection do not always work
- Know different categories of selection methods

# MOTIVATING EXAMPLE 1: REGULARIZATION

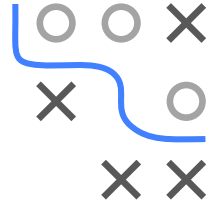


In case of $p \gg n$, overfitting becomes increasingly problematic, as can be shown by the following simulation study:

- For each of 100 simulation iterations:
- Simulate 3 datasets with $n = 100$, $p \in \{20, 100, 1000\}$.
- Features are drawn from a standard Gaussian with pairwise correlation $\rho = 0.2$.
- Target is simulated as $y = \sum_{j=1}^{p} x_j \theta_j + \sigma\varepsilon$, where $\varepsilon$ and $\boldsymbol{\theta}$ are both sampled from standard Gaussians, and $\sigma$ is fixed such that the signal-to-noise ratio is $\text{Var}(\mathbb{E}[y|X])/\sigma^2 = 2$.
- Three ridge regression models with $\lambda \in \{0.001, 100, 1000\}$ are fitted to each simulated dataset.

# MOTIVATING EX. 2: COMPARISON OF METHODS

Generalization performance of eight classification methods on micro-array data with $|\mathcal{D}_{\text{train}}| = 144$, $|\mathcal{D}_{\text{test}}| = 54$, $p = 16,063$ genes and a categorical target encoding the type of cancer with 14 classes.

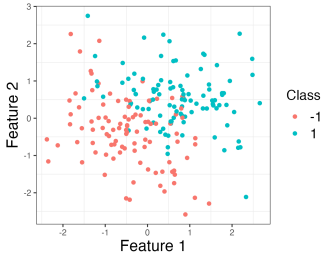


| Methods | CV errors (SE) Out of 144 | Test errors Out of 54 | Number of Genes Used |
|---|---|---|---|
| 1. Nearest shrunken centroids | 35 (5.0) | 17 | 6,520 |
| 2. $L_2$-penalized discriminant analysis | 25 (4.1) | 12 | 16,063 |
| 3. Support vector classifier | 26 (4.2) | 14 | 16,063 |
| 4. Lasso regression (one vs all) | 30.7 (1.8) | 12.5 | 1,429 |
| 5. $k$-nearest neighbors | 41 (4.6) | 26 | 16,063 |
| 6. $L_2$-penalized multinomial | 26 (4.2) | 15 | 16,063 |
| 7. $L_1$-penalized multinomial | 17 (2.8) | 13 | 269 |
| 8. Elastic-net penalized multinomial | 22 (3.7) | 11.8 | 384 |

Hastie (2009). The Elements of Statistical Learning

Methods need at least regularization or built-in FS to perform well.
Possible to build good, small models which helps in interpretation

# MOTIVATING EX. 3: INTEGRATED SELECTION

**Set-up for simulated micro-array data:**

- We generate $n = 200$ samples with $p = 100$ features drawn from a MV Gaussian
- 50% are relevant for the target and 50% have no influence
- Among informative features, 25 are positively and 25 negatively correlated with target, using weights $\{-1, 1\}$
- Target is simulated from Bernoulli distribution using linear predictor as log-odds (linear decision boundary!)